

Admissible heuristics in A*

Notation

A **node** stores a state together with a particular path from the root to that state. In tree search, the same state may appear in more than one node. In A*, the following information is maintained about each node v :

- $g(v)$ is the cost of the path found by the search from the root to a state
- $h(v)$ is the heuristic estimation of the cost from the state represented by v and the closest goal
- $f(v) = g(v) + h(v)$ is the estimated cost of reaching the goal through a path that passes through v
- A* will also typically maintain the parent of each node in order to be able to reconstruct paths

Additionally, we will use the following notation:

- $e(v, u)$ represents the single edge cost of the directed edge between the states represented by v and u
- $p^*(v, u)$ represents the optimal cost of any path between the two states

Algorithm

A* works by initializing the *fringe* with the root node, and iteratively removing from the fringe the node with the smallest f value. *Expanding* the selected node means adding all of its successors to the fringe.

If a node u is created while expanding v , then its cost is calculated as such:

- $g(u) = g(v) + e(v, u)$
- $f(u) = g(u) + h(u)$
- $parent(u) = v$

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Pseudocode:

None

A-STAR(root):

```
g(root) <- 0
f(root) <- h(root)
parent(root) <- null

fringe <- priority queue containing root (ordered by f(v))

while fringe is not empty:
    v <- extract-min(fringe)

    if v is a goal state:
        return reconstruct_path(v)

    for each successor u of v:
        parent(u) <- v
        g(u) <- g(v) + e(v, u)
        f(u) <- g(u) + h(u)
        insert u into fringe

return failure
```

Admissible heuristics

A heuristic is called **admissible** if it never overestimates the cost of the optimal path from a state to any goal state. In other words, for an admissible heuristic

$$h(v) \leq p^*(v, t) \text{ for every goal state } t \text{ reachable from } v$$

We will further assume that h is non negative, that is, $h \geq 0$ for all states

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Check your understanding:

Question 1: Suppose the fringe contains the following two nodes

- v with $g(v) = 10$, $h(v) = 5$
- u with $g(u) = 7$, $h(u) = 6$

Compute their respective f costs. Which node will A* expand first?

Question 2: Suppose that, while expanding v , $g(v) = 10$, and it has a successor state w with $e(v, w) = 5$, $h(w) = 3$. Compute $g(w)$, $f(w)$

Question 3: If h is admissible and non-negative, and t is any goal state, what value must $h(t)$ be?

Question 4: Suppose $h=0$ for all nodes. Is h admissible? What algorithm does A* become equivalent to with this heuristic?

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Optimality of A* with admissible heuristics

Theorem: If h is admissible, then tree-search A* returns a path with optimal (minimum) cost among all paths from the root to any goal state.

Note: This proof holds for tree-search A*. For graph-search A* we either need consistency, a stronger condition than admissibility, or taking proper care to reopen nodes when needed.

Proof: Suppose A* found a path to a goal state v , with total cost C , and u is the parent of v in the returned path.

Question 5: What is the cost of v in the fringe in terms of $g(u)$ and $e(u,v)$? How does it relate to C ?

Suppose v is about to be expanded. That is, $f(v)$ is minimum among all vertices in the fringe.

Assume, by contradiction, that the path returned by A* is not optimal. This must mean there is a path to another goal state, v' , with corresponding path cost $C' < C$.

Question 6: let w be the first not-yet-expanded node on the optimal root to v' path. Why must the fringe contain such a state?.

Since w is in the fringe, it must have $f(w) = g(w) + h(w)$

But because h is admissible, we must have $h(w) \leq p^*(w, v)$

Question 7: state the cost of the optimal path from root to v' , C' , in terms of $g(w)$, $h(w)$ and/or $p^*(w, v')$

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So far, we have the following:

1. $C' < C$ by assumption that A^* failed to find the optimal path
2. $f(v) = g(v) = g(u) + e(u, v) = C$ by construction
3. $f(w) = g(w) + h(w)$ by construction
4. $h(w) \leq p^*(w, v')$ because h is admissible
5. $C' = g(w) + p^*(w, v')$ by construction
6. $f(w) \leq C'$ from (3,4,5)
7. $f(w) < C$ from (6,1)
8. $f(w) < f(v)$ from (7,2)

This contradicts the fact that $f(v)$ is the minimum cost on the fringe (or equivalently, w should have been expanded before v , so w is not the first non-expanded node in the path to c'). Therefore, C is indeed the cost of the optimal path and A^* is optimal under an admissible heuristic

Question 8: Show that A^* may fail to find the optimal path under a non-admissible heuristic by drawing a tree and choosing appropriate edge weights and node heuristic values.